Multiscale modelling of strian dependent conductivity in Piezoresistive materials

Shah Yusuf Safi

April 17, 2026

Abstract

Many sensing devices rely on transduction mechanisms that convert mechanical deformation into measurable electrical signals. A prominent example is the piezoresistive effect, where the electrical conductivity of a material changes under mechanical strain.

This phenomenon originates from modifications in the underlying electronic transport properties, which can be influenced by lattice deformation, carrier mobility variations, and, in certain regimes, tunneling effects.

This thesis aims to develop a computational framework for modeling strain-dependent conductivity in deformable materials by connecting physical mechanisms at different scales. At the continuum level, the mechanical behavior is described by the equations of elasticity, while the electrical response is governed by a conductivity equation with strain-dependent coefficients. This leads to a coupled system of partial differential equations of the form

$$\nabla \cdot \boldsymbol{\sigma}(u) = 0, \qquad \nabla \cdot (\sigma(\varepsilon(u))\nabla V) = 0$$

,where u denotes the displacement field, V the electrical potential, and $\sigma(\varepsilon(u))$ represents an effective conductivity capturing the influence of mechanical deformation.

The conductivity function is treated as an effective quantity, allowing for future incorporation of transport models derived from electronic structure calculations.

Rather than treating this dependence purely phenomenologically, the work discusses how such constitutive relations can be interpreted in terms of underlying electronic transport mechanisms, providing a physically motivated link between microscopic behavior and macroscopic modeling.

From a computational perspective, the study focuses on the formulation and numerical solution of the coupled system, with particular emphasis on flexible representations of the strain–conductivity relationship. This includes the exploration of hybrid approaches that combine physics-based modeling with data-driven approximations.

The proposed framework aims to provide a consistent and extensible approach for studying electromechanical coupling in materials, with potential applications in sensing technologies, flexible electronics, and structural health monitoring

Contents

1 Introduction **2**

2 Problem Statement and Motivation **2**

3 Mathematical Model **4**

4 State of Research **5**

5 Methodology **6**

6 Expected Outcome **7**

1 Introduction

The piezoresistive effect arises from the coupling between mechanical deformation and electrical conduction. In typical applications, a doped semiconductor element (such as silicon) is embedded within a mechanical structure, where external forces induce strain in the material. This deformation alters the electronic transport properties of the material, leading to measurable changes in its electrical response, commonly observed as variations in voltage or resistance.

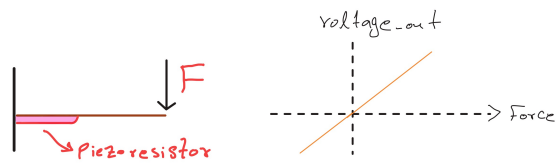

Figure 1: A schematic representation of the piezoresistive effect, where mechanical deformation leads to changes in electrical response.

The piezoresistive effect, first observed by William Thomson (Lord Kelvin) in the 19th century and later developed for semiconductors by Bardeen, Shockley, and Smith [3], describes the dependence of electrical conductivity on mechanical deformation. At a microscopic level, this behavior can be attributed to changes in the electronic structure and carrier transport mechanisms induced by strain, which modify quantities such as mobility and effective mass.

There exists a wide spectrum of engineering realizations of the piezoresistive effect, including strain gauges, pressure sensors, microelectromechanical systems (MEMS), cantilever-based sensing devices, and flexible or wearable systems [1]. Although these systems differ in geometry, material composition, and boundary conditions, they share a common physical origin rooted in the interaction between mechanical deformation and electronic transport.

Despite extensive technological development and optimization, the modeling of this coupling is often performed in a phenomenological and application-specific manner, where the dependence of conductivity on strain is introduced empirically. From a computational materials perspective, this raises the question of how such macroscopic descriptions can be related to the underlying physical mechanisms governing charge transport.

This work aims to address this by developing a computational framework that captures the electromechanical coupling at the continuum level, while maintaining a clear connection to the underlying physical processes. In this context, the continuum model serves as an effective description of strain-dependent conductivity, providing a bridge between microscopic transport phenomena and macroscopic device behavior.

2 Problem Statement and Motivation

Consider a conductive element with length L, cross-sectional area A, and resistivity ρ. Its electrical resistance is given by

$$R = \frac{\rho L}{A}. \tag{1}$$

Under mechanical loading, both the geometry and the intrinsic material properties of the conductor may change. In particular, the length, cross-sectional area, and resistivity become functions of deformation, making the resistance dependent on the mechanical state.

Using Ohm's law,

$$R = \frac{V}{I}, \tag{2}$$

and expressing voltage and current in terms of field quantities,

$$V = EL, \qquad I = JA, \tag{3}$$

we obtain

$$\frac{EL}{JA} = \frac{\rho L}{A}. \tag{4}$$

Canceling the geometric factors leads to the local relation

$$\mathbf{J} = \sigma \mathbf{E}, \qquad \sigma = \frac{1}{\rho}, \tag{5}$$

where $\mathbf{E} = -\nabla V$ is the electric field. This formulation expresses Ohm's law in a local (continuum) form, independent of the specific geometry.

Combining this with charge conservation in the steady state,

$$\nabla \cdot \mathbf{J} = 0, \tag{6}$$

yields the conductivity equation

$$\nabla \cdot (\sigma \nabla V) = 0. \tag{7}$$

This equation provides a continuum description of electrical conduction in materials, where the conductivity σ may vary spatially. It belongs to the class of elliptic partial differential equations in divergence form [2].

In piezoresistive materials, the key feature is that the conductivity is no longer constant, but depends on the mechanical deformation of the material. At a macroscopic level, this dependence can be expressed as

$$\sigma = \sigma(\varepsilon(u)), \tag{8}$$

where the strain tensor is given by[1]

$$\varepsilon(u) = \frac{1}{2}(\nabla u + \nabla u^T). \tag{9}$$

This relation can be interpreted as an effective description of how mechanical deformation influences electrical transport. Physically, the dependence of conductivity on strain originates from

[1]The strain tensor is defined as the symmetric part of the displacement gradient, which removes rigid body rotations and captures the true deformation of the material.

changes in the underlying electronic structure and carrier transport properties, which are affected by lattice deformation.

The resulting model therefore introduces a nonlinear coupling between mechanical deformation and electrical conduction, linking continuum mechanics with electronic transport in a unified framework.

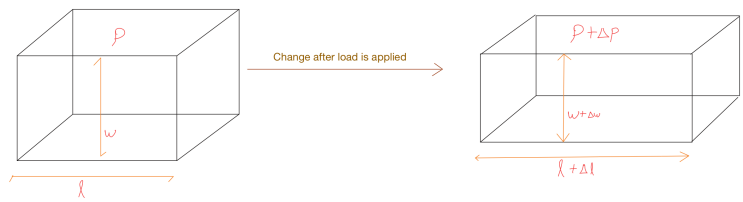

Figure 2: Change of geometric and material properties under mechanical deformation. L denotes the length of the conductor, W its characteristic width, and $A = W^2$ the cross-sectional area (assuming a square cross-section). The material property ρ represents the electrical resistivity, which may vary under mechanical deformation. Both geometric changes and intrinsic material variations contribute to the overall change in resistance.

This illustrates that both geometric deformation and intrinsic material changes contribute to the variation of resistance, providing the basis for modeling the piezoresistive effect at the continuum level.

3 Mathematical Model

Based on the physical considerations discussed above, the coupled electromechanical behavior can be described at the continuum level by a system of partial differential equations.

The mechanical response of the material is governed by the equations of elasticity,

$$\nabla \cdot \boldsymbol{\sigma}(u) = 0, \tag{10}$$

where u denotes the displacement field and $\boldsymbol{\sigma}(u)$ is the stress tensor, which depends on the strain.

The electrical response is described by the conductivity equation,

$$\nabla \cdot (\sigma(\varepsilon(u))\nabla V) = 0, \tag{11}$$

where V is the electrical potential and the conductivity σ depends on the strain tensor $\varepsilon(u)$.

Together, these equations define a coupled system in which mechanical deformation influences electrical conduction through the strain-dependent conductivity. From a physical perspective, this coupling represents an effective description of how deformation alters the underlying electronic transport properties of the material.

This formulation provides the basis for computational modeling of the piezoresistive effect within a continuum framework.

4 State of Research

The piezoresistive effect has been extensively investigated in both experimental and engineering contexts, particularly in the development of semiconductor-based sensors and microelectromechanical systems (MEMS) [3, 1]. Early studies established the strong dependence of electrical conductivity on mechanical strain in semiconductors such as silicon, highlighting the role of crystal structure and carrier transport in determining the sensitivity of the effect.

From a materials physics perspective, piezoresistivity originates from strain-induced modifications of the electronic structure and charge transport properties. In semiconductors, mechanical deformation can alter band structure, carrier mobility, and scattering mechanisms, leading to significant changes in conductivity. These effects have been studied using both phenomenological models and more detailed electronic transport theories.

In practical applications, however, the coupling between mechanical deformation and electrical response is often modeled in a simplified and application-specific manner, where the dependence of conductivity on strain is introduced empirically. While such approaches are effective for device design and optimization, they do not always provide a transparent connection between the underlying physical mechanisms and the resulting macroscopic behavior.

At the continuum level, models based on partial differential equations are widely used to describe both elasticity and electrical conduction. However, the integration of strain-dependent transport properties into such models is typically treated in a phenomenological way, and a systematic computational framework that bridges physical insight and continuum modeling remains an open challenge.

Such a framework is particularly relevant for understanding emerging materials and devices where electromechanical coupling plays a central role.

Based on these considerations, the proposed study is guided by the following research questions:

- How can strain-dependent conductivity be incorporated into a consistent continuum model of electromechanical coupling?
- How does mechanical deformation influence the spatial distribution of electrical potential in conductive materials?
- What qualitative and quantitative effects arise from the nonlinear coupling between mechanics and electrical transport?
- To what extent can flexible (e.g. data-driven or physics-informed) approaches improve the modeling of $\sigma(\varepsilon)$?

To address these questions, the research will be structured around the following key components:

- Physical background and modeling assumptions
- Formulation of the coupled electromechanical model
- Analysis of the nonlinear system
- Computational methods and numerical experiments
- Interpretation of results in the context of material behavior

‘

5 Methodology

The proposed study follows a computational approach based on a continuum description of the coupled electromechanical system. The methodology is structured in several stages, combining physical modeling with numerical approximation.

First, the governing equations of elasticity and electrical conduction are formulated within a consistent continuum framework. The mechanical problem provides the displacement field u, from which the strain tensor $\varepsilon(u)$ is computed. This strain field serves as input to the electrical problem through the strain-dependent conductivity $\sigma(\varepsilon)$.

As a baseline, the electrical conduction problem with constant conductivity is considered and solved numerically. This case admits an analytical solution and is used to validate the numerical implementation. A finite element discretization is employed to approximate the solution, and the numerical results are compared against the exact solution.

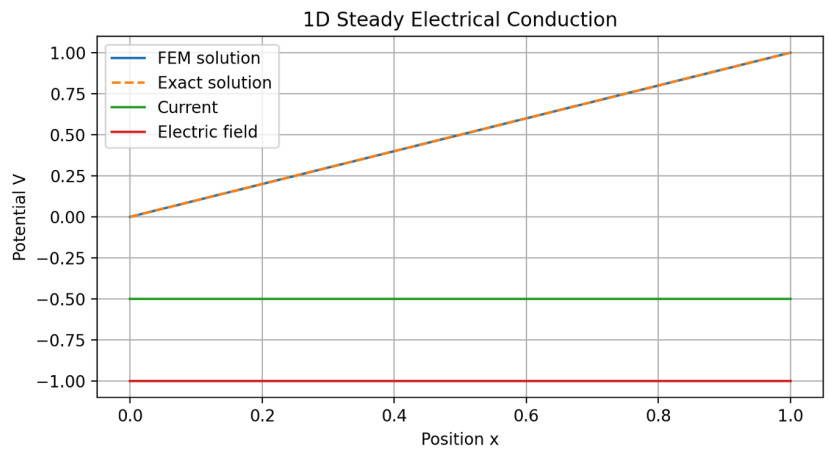

Figure 3: Finite element solution of the 1D steady conduction problem with constant conductivity compared with the analytical solution. The linear potential profile and constant electric field confirm the expected physical behavior.

Building on this, the full coupled system is addressed by introducing a spatially varying conductivity $\sigma(\varepsilon(u))$. The problem is solved iteratively: the mechanical and electrical fields are updated in a coupled manner until convergence is achieved. Special attention is given to the nonlinear nature of the coupling, which may lead to non-uniform fields and sensitivity to deformation.

From a computational perspective, the finite element method (FEM) is used as the primary discretization technique due to its flexibility in handling heterogeneous material properties and complex geometries. The implementation focuses on robustness and clarity, enabling systematic exploration of different forms of the strain–conductivity relation.

In addition, the methodology allows for extensions beyond purely phenomenological models. In particular, the constitutive relation $\sigma(\varepsilon)$ can be treated as a flexible component, which may be informed by physical insight or approximated using data-driven approaches. This provides a pathway to incorporate more detailed transport behavior without modifying the overall computational framework.

Overall, the methodology is designed to provide a transparent and extensible computational pipeline, bridging continuum modeling with physically motivated descriptions of strain-dependent conductivity.'

6 Expected Outcome

The expected outcome of this study is the development of a computational framework for analyzing strain-dependent electrical conduction in deformable materials. This framework will integrate continuum mechanics and electrical transport into a unified numerical model, allowing for the systematic investigation of electromechanical coupling.

At the modeling level, the work will provide a consistent formulation of conductivity as a strain-dependent quantity, together with a numerical implementation capable of solving the resulting coupled system. This will enable the exploration of how mechanical deformation influences electrical potential distribution, current density, and overall sensitivity of the system.

From a computational perspective, the study will deliver a flexible pipeline that can be adapted to different forms of the strain–conductivity relation, including both phenomenological models and more physically informed or data-driven descriptions. This makes the framework suitable for studying a range of materials and configurations.

Beyond the specific application, the results are expected to contribute to the broader understanding of nonlinear multiphysics systems, particularly those involving coupling between mechanical deformation and transport phenomena. Potential applications include piezoresistive sensing, structural health monitoring, and the design of flexible or functional materials.

The framework may also serve as a basis for future extensions toward multiscale or more detailed transport models.

References

[1] Joseph C. Doll and Beth L. Pruitt. *Piezoresistor Design and Applications*. 2013.

[2] Lawrence C. Evans. *Partial Differential Equations*. 2010.

[3] C. S. Smith. "Piezoresistance effect in germanium and silicon". In: *Physical Review* 94.1 (1954), p. 42.